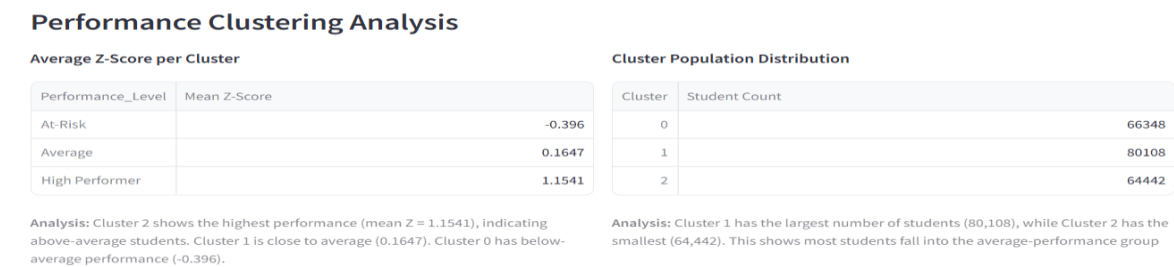


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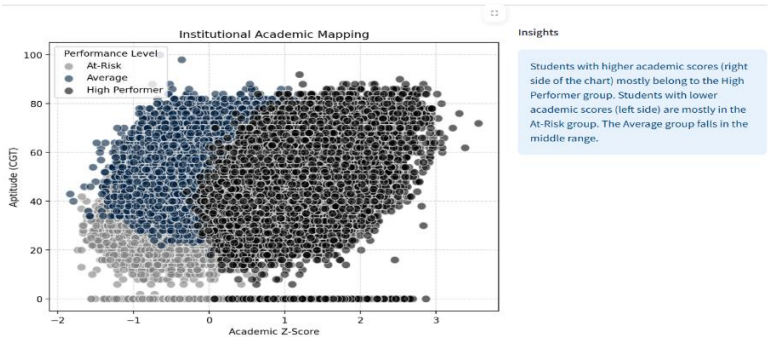
Individual Project Contribution Report: G.C.E A/L 2020 Performance Analytics

Our group analyzed G.C.E A/L 2020 student performance, primarily focusing on Z-score prediction. As my individual contribution, I introduced an advanced analytical extension using unsupervised learning. I implemented a MiniBatch K-Means clustering model. I chose MiniBatch K-Means over Standard K-Means as the dataset contains over 200000 records (to improve computational efficiency).

Visualization: Cluster- wise average Z scores and student sample distribution across clusters



Visualization: Scatterplot of Z score VS CGT (2D academic performance mapping)



An interactive module that predicts the Z score (from the group project), classifies the students into a performance tier, and provides strategic academic recommendations (from my individual contribution).

